



Characteristics of aerodynamic forces on high-rise buildings with various façade appurtenances

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ARTICLE INFO

Keywords:

High-rise building
Façade appurtenances
Wind tunnel experiment
Wind loads
Correlation coefficients

ABSTRACT

The appurtenances effects of high-rise building on wind loads are investigated through a series of wind tunnel experiments with 21 façade appurtenances simulated by different configurations of horizontal splitter plates. “Zigzag variations” of layer forces can be found in both along-wind and cross-wind when the vertical separation distance ratio of continuous plates is larger than 4%. The appurtenances can significantly reduce fluctuating pressures on the side façade but with different effect on the fluctuating cross-wind layer force. The reason of this will be discussed in this study. The power spectra of response of the building model indicate that the horizontal façade appurtenances have negligible effects on the period of vortex shedding. But the peaks in the power spectra become slightly higher than those from a smooth surface model, and the bandwidths are also narrower. Such results indicate that the horizontal plates may result in better organization of the vortex shedding by restricting the flow in vertical direction and reduce the intensity of vortex. Further check on the correlation coefficients between pressures at the same level reviews that the correlation of response from different positions of the building models with appurtenances is higher than that for building model with smooth surface. It is also noted that discontinuous appurtenances can reduce the cross-wind base moment by approximately 5%.

1. Introduction

High-rise buildings are becoming slender and more flexible with the increase of height. Wind load, thus, is more critical for the design of high-rise buildings. Many factors can significantly affect the wind load acting on high-rise buildings, such as the up wind terrain, appearance of neighboring building (Yu et al., 2018; Hui et al., 2017), building shape (Kwok et al., 1988), etc.

Many researchers have focused on aerodynamic treatments of building shape to improve the aerodynamic performance. Most of them tried to reduce the overall wind load especially the dynamic cross-wind load through altering or suppressing the generation of vortex. The treatments include corner cut, chamfer, taper, set-back, openings and helical (Carassale et al., 2014; Li et al., 2018). Tamura et al. (2017) investigated the aerodynamic characteristics of super-tall building models with various unconventional configurations based on a series of comprehensive experimental studies. They reported that most configurations can reduce aerodynamic force, but helical models and polygon models are better. It has been proved that these approaches could reduce the wind-induced response and aerodynamic force effectively.

Some studies have tried to reduce the local wind load for cladding design through local small scale changes, which scarcely alter the shape of building. It is another kind of effective aerodynamic treatment to optimize the aerodynamic characteristics. Stathopoulos and Zhu (1988, 1990) investigated the effect of uniform roughness, such as balconies and mullions, on the wind pressure. They reported that balconies with a width up to 4m without walls on a 120m high square building could reduce slightly the pressure coefficient on the building surface, whereas mullions could induce adverse wind effects at wall edges. They also made suggestions to the design of balcony based on their study. Maruta et al. (1998) reported on the effects of width of balcony on the local wind pressure with the balconies fixed on the façade. The rate of reduction of peak pressure is 0.67 when the surface roughness magnitude equals 1/10. The role of turbulence and the mechanism of local peak pressure reduction effect were also discussed. Kumar et al. (2013) investigated the effect of corner balcony on wind-induced response of tall building, and reported that the extended out corner balcony could reduce the cross-wind base loads in the range of 10–30%. Quan et al. (2017) investigated the effect of vertical ribs protruding from building façade on wind loads and reported that the vertical ribs significantly reduce the

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<https://doi.org/10.1016/j.jweia.2019.06.002>

Received 8 April 2019; Received in revised form 17 May 2019; Accepted 1 June 2019

Available online 8 June 2019

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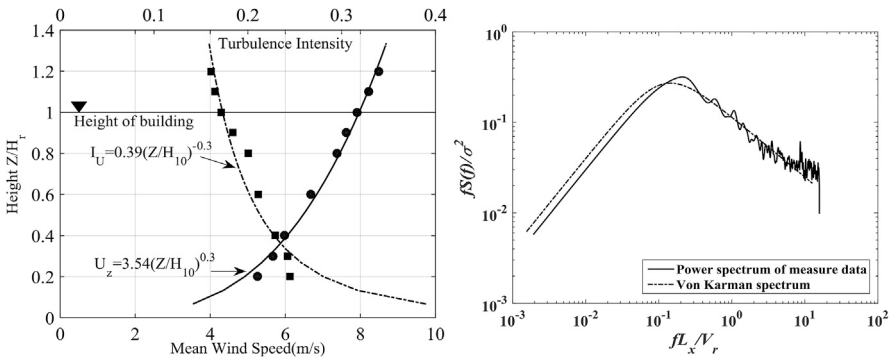
Fig. 1. Wind tunnel setup and models.

most unfavorable suction coefficients in the corner recession regions and increase the mean and overall fluctuating along-wind aerodynamic forces. Yuan et al. (2018) investigated the local wind pressure on high-rise buildings basing on the experiment with 21 configurations of horizontal plates. It was found that the largest negative peak pressure at the higher leading corner can be reduced by 42% from the case with smooth surface condition. More experiment investigations on facade appurtenances effect on low-rise buildings or high-rise

buildings with circular or ellipse section were also carried out (Chand et al., 1998; Letchford et al., 2016; Quan et al., 2016). Past researches on local small scale changes have mainly focused on the wind pressure of the building façade. However, their effects on layer force and base moment may also need to be carefully considered. This study investigates the effects of façade appurtenances on the wind loads of high-rise buildings. Wind tunnel experiments were carried out based on a prototype of smooth façade building. Buildings with the same cross-sections but different types of appurtenance are adopted for the investigation. Only the horizontal thin plates were adopted as the appurtenance in this study, so that the effects of the change of shape of the cross-section are avoided. Base moment coefficients and layer force coefficients in the along-wind and cross-wind directions are checked, respectively.

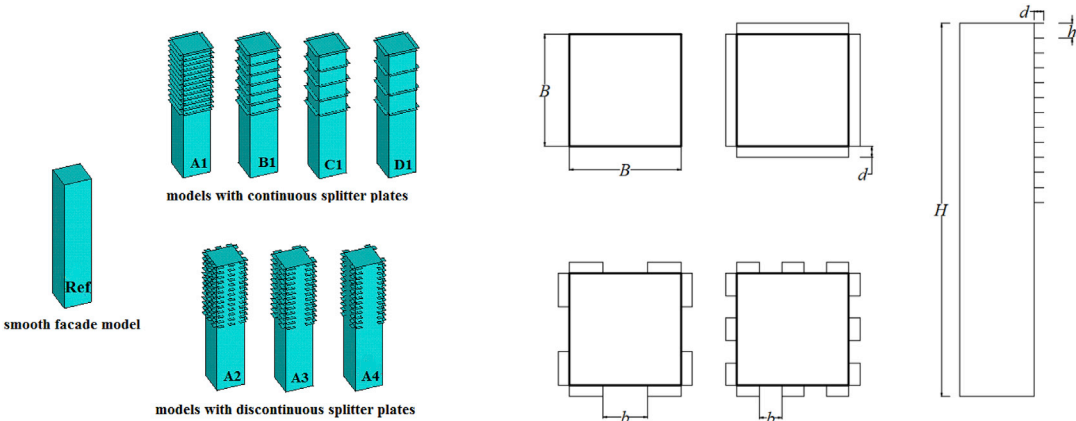
Table 1
Parameters of the appendant.

Category	h	$h_r = h/H$	b	$b_r = b/B$
Ref	/	/	/	/
A1	20 mm	4%	/	/
A2	20 mm	4%	20 mm	20%
A3	20 mm	4%	40 mm	40%
A4	20 mm	4%	60 mm	60%
B1	40 mm	8%	/	/
C1	60 mm	12%	/	/
D1	80 mm	16%	/	/



(a) Profile of mean wind speed and turbulence intensity (b) Power spectrum of fluctuating wind speed

Fig. 2. Wind field information and characteristic in this study.



(a) Three-dimensional sketch of models (b) Model configurations and appurtenances parameters

Fig. 3. Model information in this study.

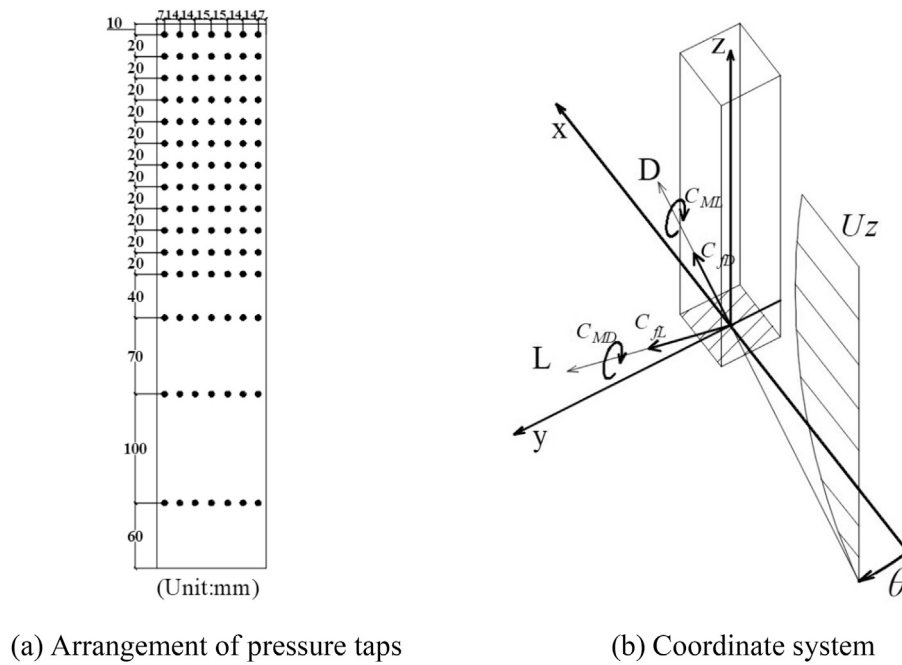
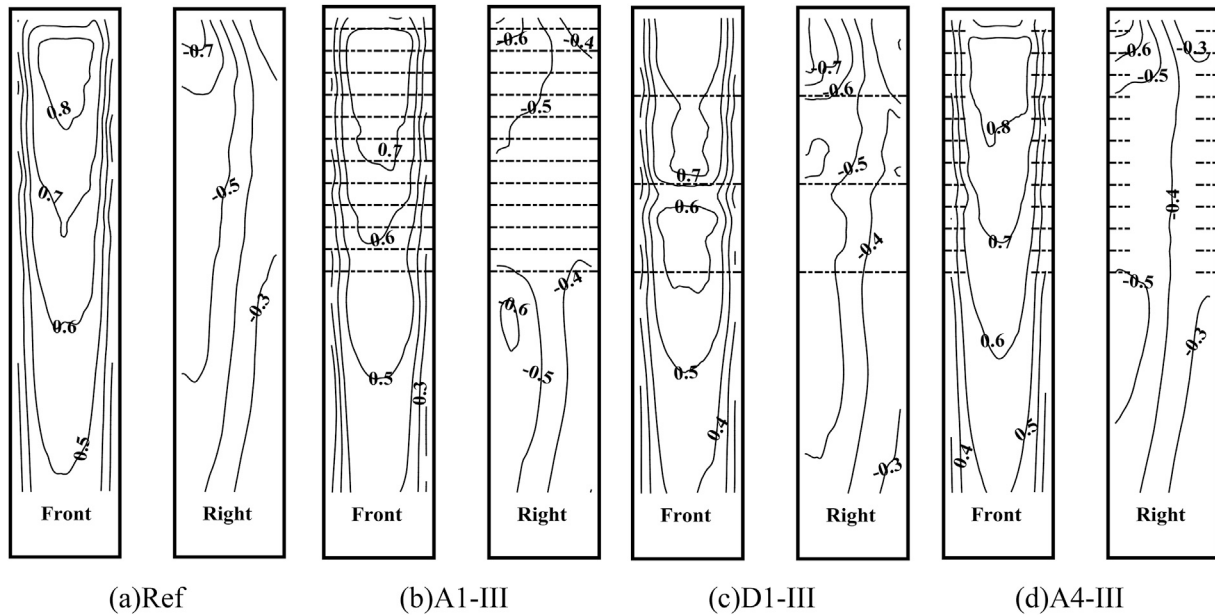


Fig. 4. Detailed experiment and model information.

Fig. 5. Distribution of $\overline{C_p}(j, 0^\circ)$.

2. Wind tunnel experiments

Wind tunnel experiments are carried out in the boundary layer wind tunnel at Hunan University, China. The test section of the wind tunnel is 3 m wide, 2.5 m high and 11.5 m long. The arrangement in wind tunnel is as shown in Fig. 1. Spires-and-roughness-blocks technique is adopted to simulate a boundary layer wind flow field of urban terrain and the power law index (α) of the mean wind speed profile is 0.3 (GB 50009-2012). The profiles of mean wind speed and turbulence intensity are shown in Fig. 2(a). A velocity scale of 1/8 is used in this study. The 8 m/s mean wind speed U_H at the roof height is chosen to be the reference velocity. Fig. 2(b) plots the measured power spectrum of wind speed near the reference height indicating a good agreement with the Von Karman spectrum.

A square-sectioned building model with an aspect ratio of 1/5 was studied with 500 mm height H and 100 mm width B . The façade of the reference model is smooth as shown in Fig. 3(a), while the other seven models have different kinds of appurtenances on the four sides of the building façade. The appurtenances plates were only installed on the upper part of the building since the associated wind loads are more critical for the structural design. The appurtenances, also called splitter plates, were simulated by very thin rigid horizontal plates and their parameters are shown in Fig. 3(b), where “ d ” indicates the extension depth of the thin plates; “ b ” means the horizontal gap between two adjacent plates and “ h ” represents the vertical distance between two adjacent plates. The geometrical scale is 1:300 and the Reynolds number in these experiments is $Re \approx 5.78 \times 10^4$, which is common for wind tunnel experiment of high-rise building. The appurtenances parameters are

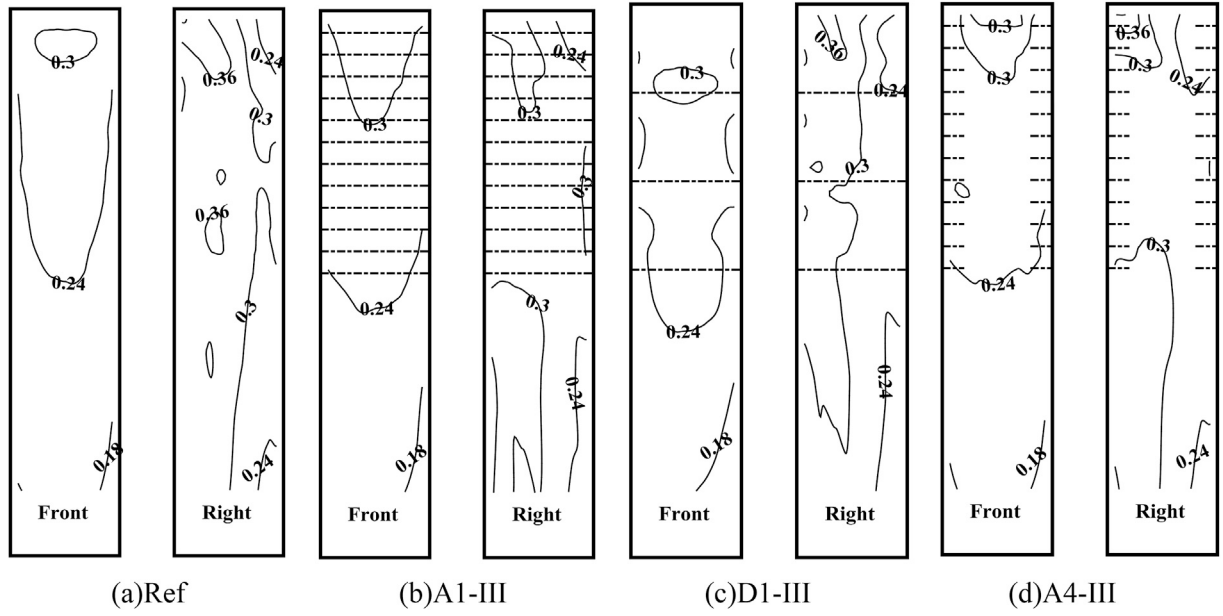
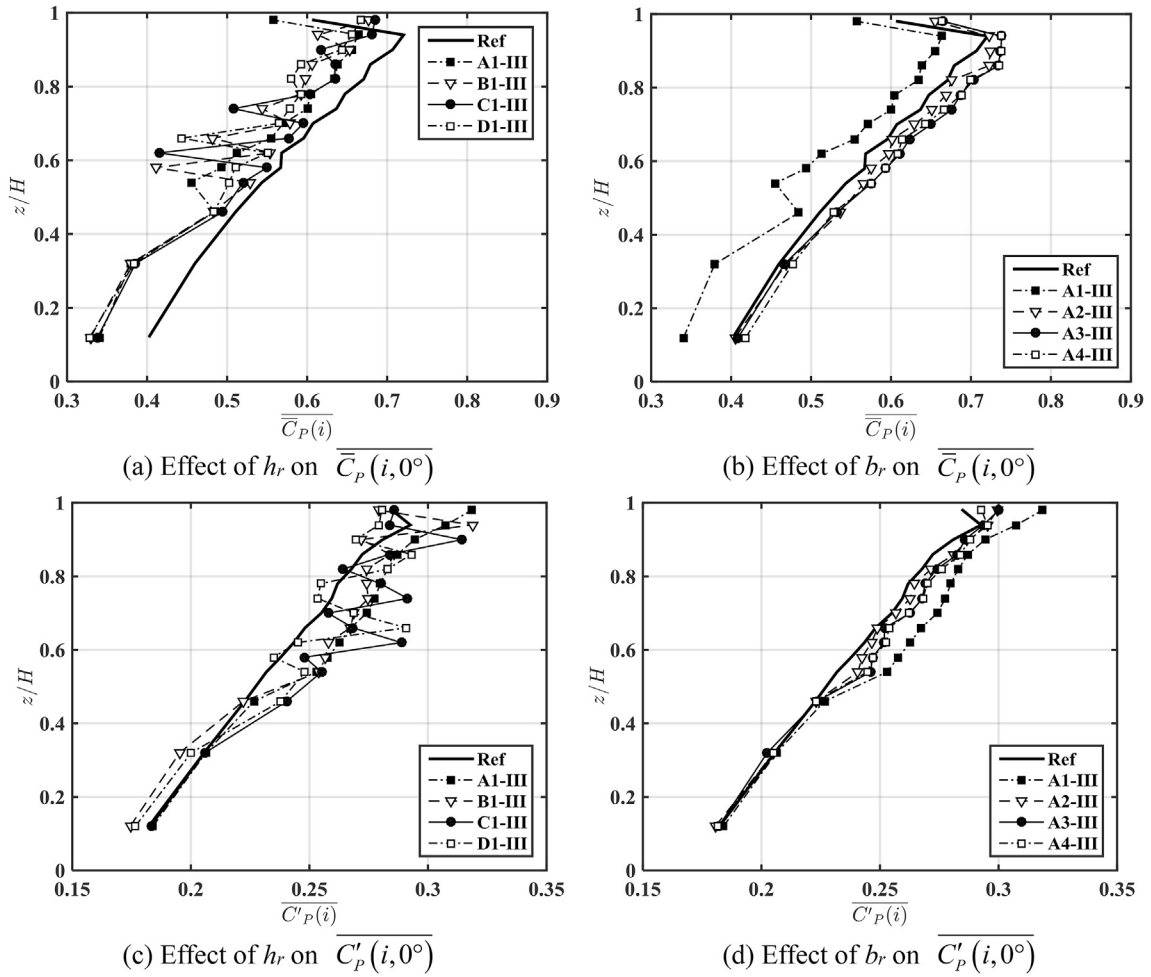
Fig. 6. Distribution of $C_p(j, 0^\circ)$.

Fig. 7. Profiles of wind pressure coefficients along height on the windward façade.

normalized into non-dimensional form (b_r and h_r), as shown in Table 1. Model Ref is the reference model, and model A1, B1, C1 and D1 have

continuous plates with different h_r , and model A2, A3 and A4 have discontinuous plates with different b_r . In addition, three different values

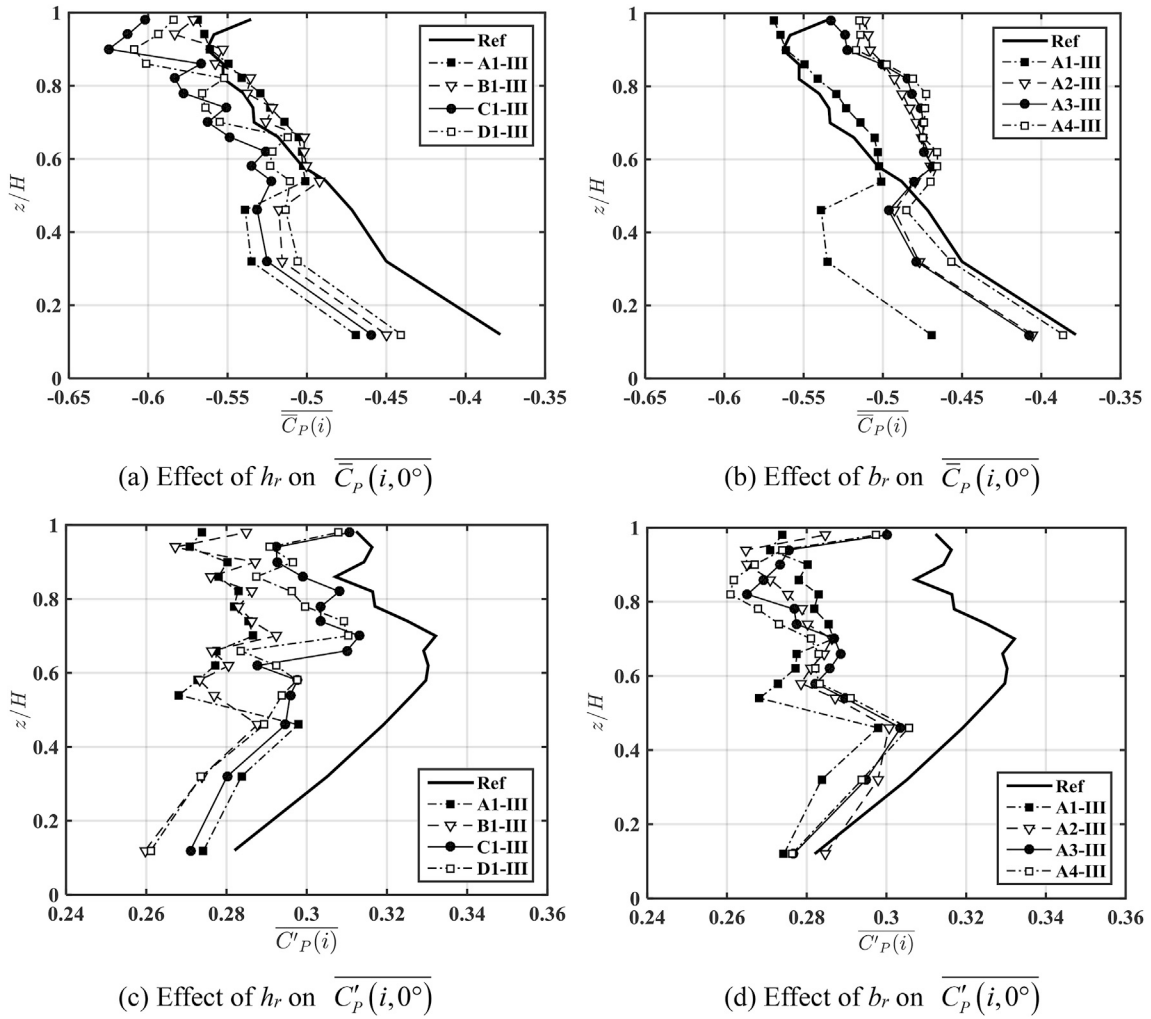


Fig. 8. Profiles of wind pressure coefficients along height on the side façade.

of d_r ($d_r = d/B$) for each appurtenances type are investigated, which are 7.5%, 10% and 12.5%, respectively. Roman numerals are used to indicate different d_r of each model, such as A1-I ($d_r = 7.5\%$), A1-II ($d_r = 10\%$) and A1-III ($d_r = 12.5\%$). Therefore, a total of 22 different models are studied.

A total of 420 pressure taps were mounted at 15 levels on the building façade, with 28 taps at each level as shown in Fig. 4(a). Fig. 4(b) shows the definitions of wind forces, moments, and the coordinate system employed in this study. The multi-channel simultaneous high frequency pressure measuring system was used to obtain the time-histories of fluctuating pressure, with a sampling frequency of 312.5 Hz. Time scale is set as 1:37.5, and fifteen 16-second long samples, corresponding to fifteen 10-minute long samples in full scale, were retained for each wind direction θ .

The measured wind pressure at each tap is expressed as a non-dimensional pressure coefficient $C_p(j, \theta)$, where j denotes the sequence number of the tap ($j = 1, 2, \dots, 420$), θ denotes the wind direction ($\theta = 0^\circ, 5^\circ, \dots, 45^\circ$). The mean and fluctuating pressure coefficients are expressed as $\overline{C_p}(j, \theta)$ and $C'_p(j, \theta)$, respectively. $\overline{C_p}(i, \theta)$ represents the average of all the mean pressure coefficients observed from the pressure taps at the i -th level in the direction of θ and $\overline{C'_p}(i, \theta)$ represents the average of all the fluctuating pressure coefficients observed from the pressure taps at the i -th level in the direction of θ .

By integrating the wind pressures from the pressure taps at each level with consideration of the tributary area of each tap, the time-histories of layer wind force and base moment can be obtained as following:

$$F_{jD}(i, \theta) = \sum_{j=1}^{28} P(j, \theta) \cdot A_j \cdot \cos \alpha \cdot \cos \theta + \sum_{j=1}^{28} P(j, \theta) \cdot A_j \cdot \sin \alpha \cdot \sin \theta \quad (1)$$

$$F_{jL}(i, \theta) = \sum_{j=1}^{28} P(j, \theta) \cdot A_j \cdot \cos \alpha \cdot \sin \theta + \sum_{j=1}^{28} P(j, \theta) \cdot A_j \cdot \sin \alpha \cdot \cos \theta \quad (2)$$

$$M_{MD}(\theta) = \sum_{i=1}^{15} F_{jD}(i, \theta) \cdot H(i) \quad (3)$$

$$M_{ML}(\theta) = \sum_{i=1}^{15} F_{jL}(i, \theta) \cdot H(i) \quad (4)$$

where $F_{jD}(i, \theta)$ and $F_{jL}(i, \theta)$ denote respectively the along-wind and cross-wind layer forces; $M_{MD}(\theta)$ and $M_{ML}(\theta)$ denote respectively the along-wind and cross-wind base moments; $P(j, \theta)$ is the wind pressure at point j at i -th level; A_j is the tributary area of point j ; α is the angle between x-axis and positive wind pressure vector of point j ; $H(i)$ is the height of i -th layer.

The normalized coefficients of layer wind force and base moment are defined as

$$\overline{C_{jD}}(i, \theta) = \frac{\overline{F_{jD}}(i, \theta)}{l(i)Bq_H} \quad C'_{jD}(i, \theta) = \frac{F'_{jD}(i, \theta)}{l(i)Bq_H} \quad (5)$$

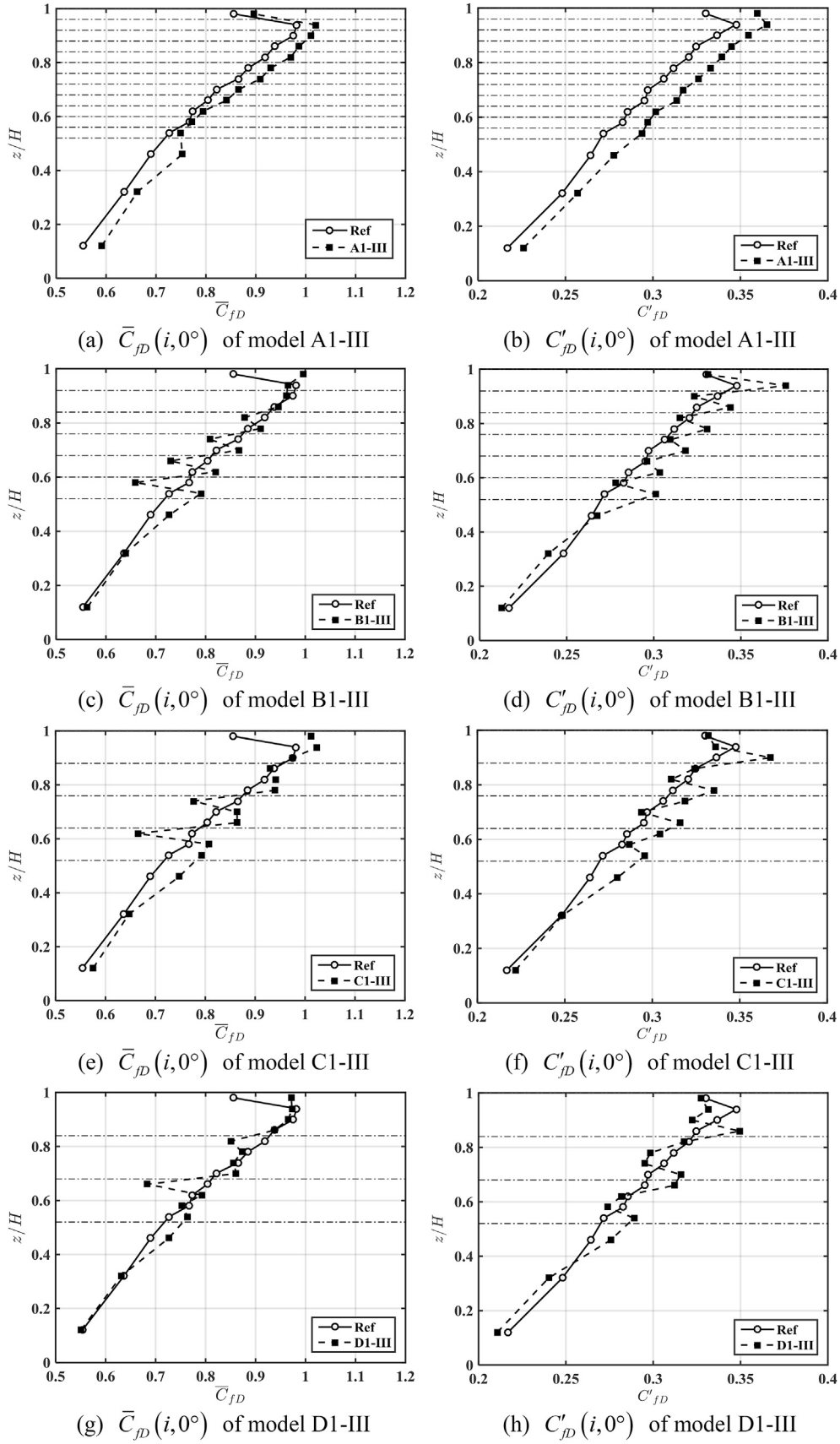


Fig. 9. $\bar{C}_{fD}(i, 0^\circ)$ and $C'_{fD}(i, 0^\circ)$ of continuous appurtenances models (different h_r).

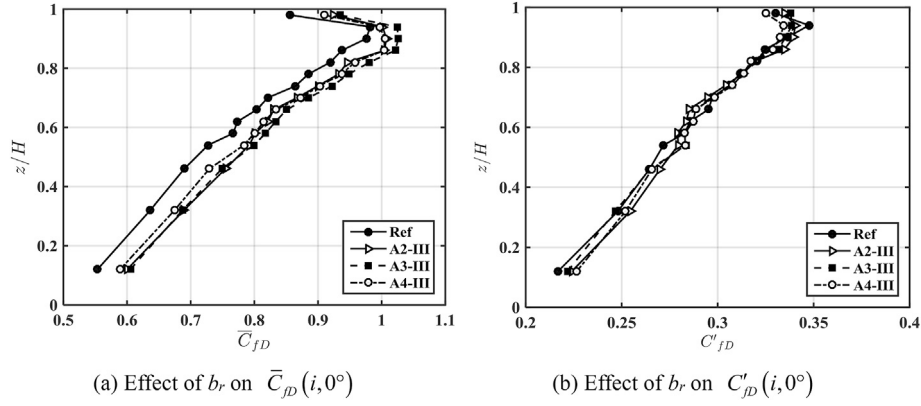


Fig. 10. $\bar{C}_{PD}(i, 0^\circ)$ and $C'_{PD}(i, 0^\circ)$ of discontinuous appurtenances models (different b_r).

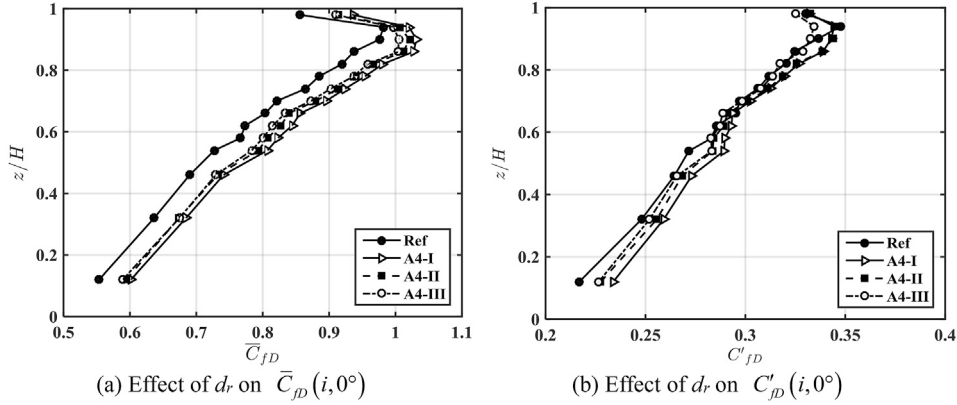


Fig. 11. $\bar{C}_{PD}(i, 0^\circ)$ and $C'_{PD}(i, 0^\circ)$ of model A4 (different d_r).

$$\bar{C}_{PL}(i, \theta) = \frac{\bar{F}_{PL}(i, \theta)}{l(i)Bq_H} \quad C'_{PL}(i, \theta) = \frac{F'_{PL}(i, \theta)}{l(i)Bq_H} \quad (6)$$

$$\bar{C}_{MD}(\theta) = \frac{\bar{M}_{MD}(\theta)}{H^2Bq_H} \quad C'_{MD}(\theta) = \frac{M'_{MD}(\theta)}{H^2Bq_H} \quad (7)$$

$$\bar{C}_{ML}(\theta) = \frac{\bar{M}_{ML}(\theta)}{H^2Bq_H} \quad C'_{ML}(\theta) = \frac{M'_{ML}(\theta)}{H^2Bq_H} \quad (8)$$

where $\bar{C}_{PD}(i, \theta)$ and $\bar{C}_{PL}(i, \theta)$ are the along-wind and cross-wind mean layer wind force coefficients at the i -th layer with wind direction θ ; correspondingly, $C'_{PD}(i, \theta)$ and $C'_{PL}(i, \theta)$ are the fluctuating layer wind force coefficients of i -th level; $\bar{C}_{MD}(\theta)$ and $\bar{C}_{ML}(\theta)$ are the mean base moment coefficients with wind direction θ ; $C'_{MD}(\theta)$ and $C'_{ML}(\theta)$ are the fluctuating base moment coefficients. $\bar{F}(i, \theta)$ and $F'(i, \theta)$ are the mean and fluctuating layer wind forces of i -th level, while $\bar{M}(\theta)$ and $M'(\theta)$ are the mean and fluctuating base bending moments; $l(i)$ is the characteristic height of i -th level; H and B are the model height and width as shown in Fig. 3; q_H denote the reference pressure at the roof height defined as $0.5\rho U_H^2$.

The $|\bar{C}_{MD}(\theta)|_{max}$, $|C'_{MD}(\theta)|_{max}$, $|\bar{C}_{ML}(\theta)|_{max}$ and $|C'_{ML}(\theta)|_{max}$ are adopted to represent the maximum values of base moment coefficients from all wind directions, and they can be calculated as

$$|C_M(\theta)|_{max} = \max_{0^\circ \leq \theta \leq 45^\circ} \{|C_M(\theta)|\} \quad (9)$$

where $C_M(\theta)$ can be either the mean or fluctuating base moment coefficient.

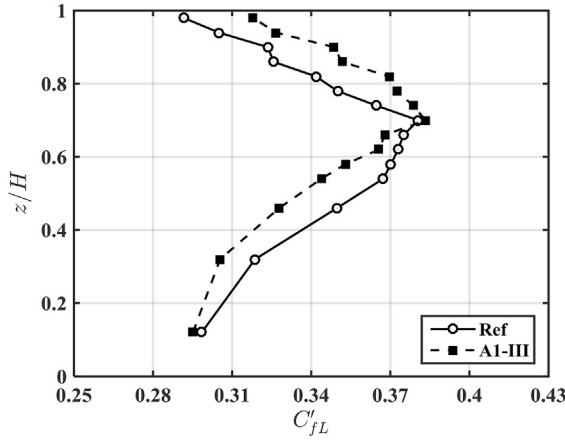
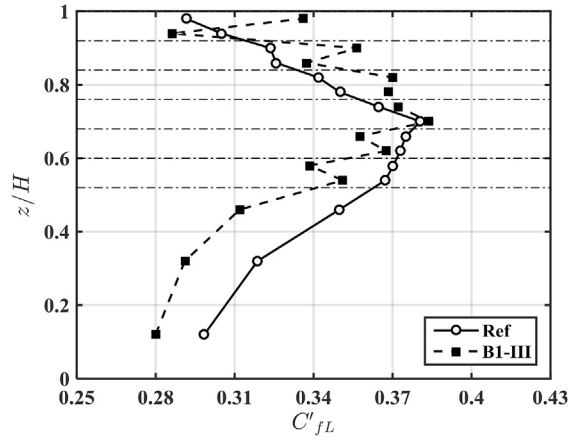
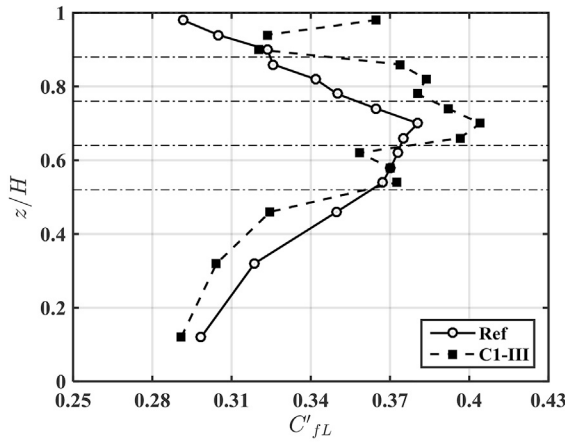
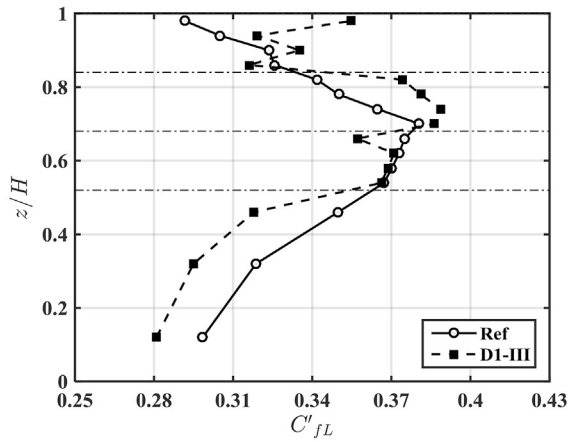
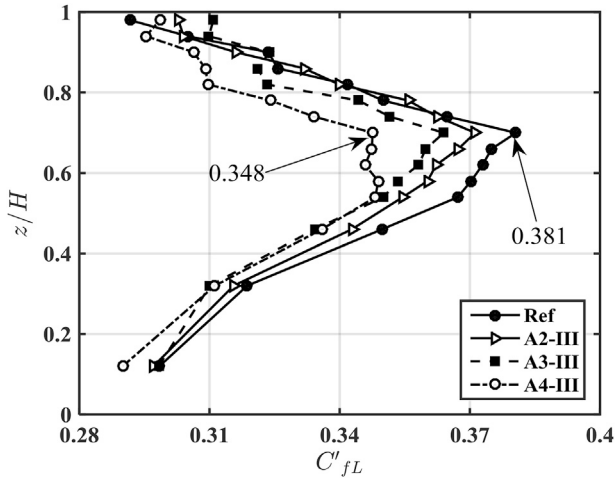
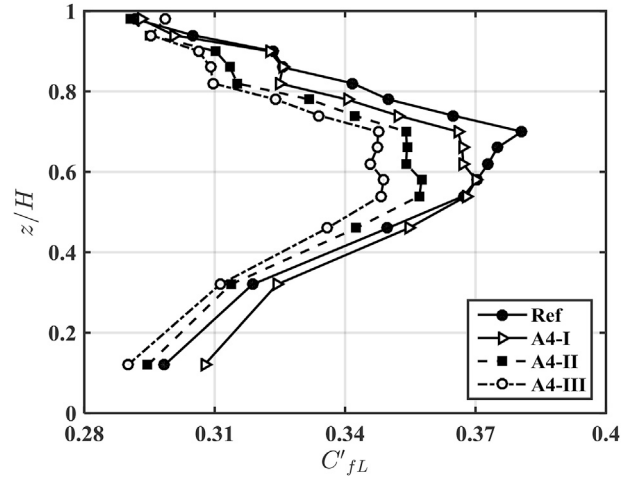
3. Experimental results and discussion

3.1. Distribution of mean and fluctuating wind pressure coefficients

Since $\theta=0^\circ$ is the most unfavorable incident wind direction for overall wind load acting on the square-sectioned building, the distribution of mean and fluctuating wind pressure coefficients at the wind direction of 0° for some of the models are shown in Figs. 5 and 6. Only those on the windward and right façade are shown in the pictures. Three types of appurtenances model will be discussed in details and compared with the model Ref in this section.

Fig. 5 shows that the maximum mean positive pressure coefficient of the model Ref is around 0.8 and locates at the $0.8H$ height of the windward wall, while the largest mean negative pressure coefficient -0.8 locates at the upper leading corner of the side face. The maximum value of $\bar{C}_P(j, 0^\circ)$ on the windward side decreases from 0.8 to 0.7 when the splitter plates are continuous or discontinuous. For models with h_r of 4%, the absolute values of $\bar{C}_P(j, 0^\circ)$ on the side face decrease at the upper leading edge when the appurtenances are continuous. It varies slightly when the appurtenances are discontinuous. However, such effects become weak when the value of h_r is large. Fig. 5(c) shows that the largest of mean negative peak pressure coefficients on the side face of model D1-III can even reach up to -0.8 which is larger than that of the model Ref. The absolute values of $\bar{C}_P(j, 0^\circ)$ at the lower part and tail edge of the side façade of the models with continuous appurtenance increase more seriously when d_r is larger. The reason of above behavior may be that the wider appurtenances strengthen the attachment of the flow on the local area under the lowest level appurtenances.

Fig. 6 shows that both continuous and discontinuous appurtenances have little effect on the distribution of $C'_P(j, 0^\circ)$ on the windward surface.

(a) $C'_{fL}(i, 0^\circ)$ of model A1-III(b) $C'_{fL}(i, 0^\circ)$ of model B1-III(c) $C'_{fL}(i, 0^\circ)$ of model C1-III(d) $C'_{fL}(i, 0^\circ)$ of model D1-IIIFig. 12. $C'_{fL}(i, 0^\circ)$ of continuous appurtenances models (different h_r).Fig. 13. $C'_{fL}(i, 0^\circ)$ of discontinuous appurtenances models (different b_r).Fig. 14. $C'_{fL}(i, 0^\circ)$ of model A4 (different d_r).

The values of $C'_p(j, 0^\circ)$ at the leading corner of the side face, however, decrease with smaller h_r , which is similar to the effect of appurtenances on the mean pressure coefficients. The largest difference from the results of model Ref is about 17% smaller for model A1-III. The $C'_p(j, 0^\circ)$ at the lower positions becomes lower for all type of model as different from the

effect on the mean pressure of the lower positions.

Fig. 7 shows the average mean and fluctuating pressure coefficients at each level on the windward façade. The $\bar{C}_p(i, 0^\circ)$ on the windward side of model Ref increases with the building height and reaches to the largest value 0.72 at $0.94H$ while the $C'_p(i, 0^\circ)$ profiles also reach the largest

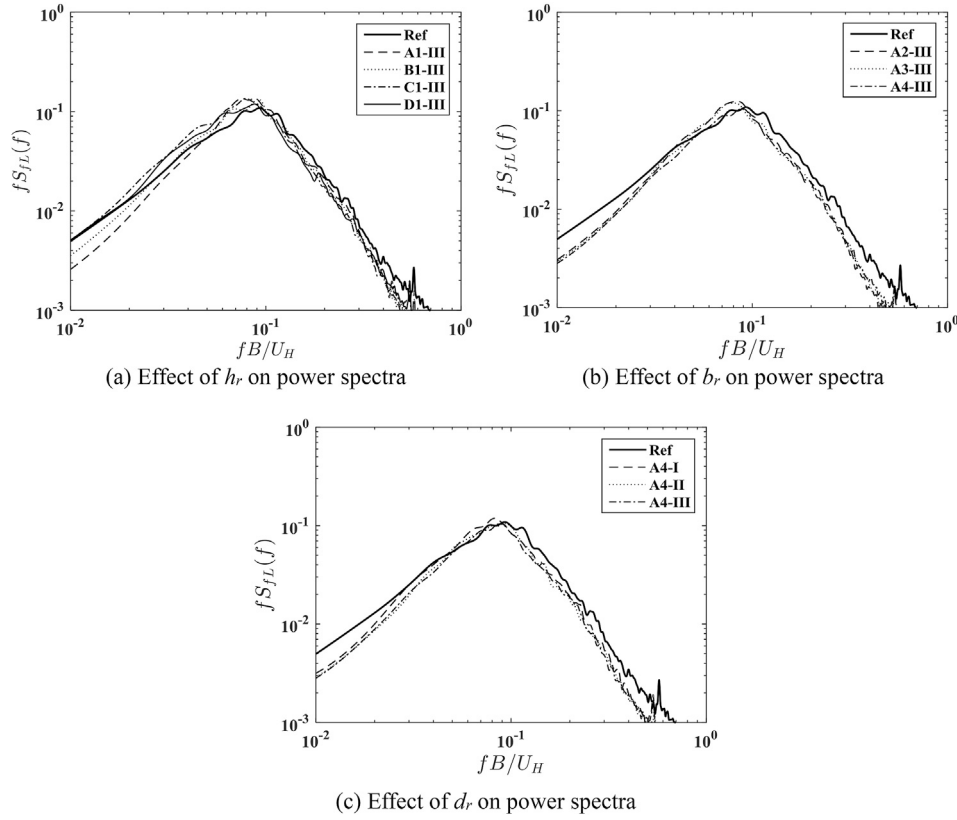


Fig. 15. Power spectra of cross-wind layer force at $0.7H$.

value 0.29 at that height. Both Fig. 7(a) and (c) show that the $\overline{C_p}(i, 0^\circ)$ on the windward side of models with continuous appurtenances are smaller than that of model Ref but the $\overline{C_p}(i, 0^\circ)$ shows larger values. It can also be noted that the $\overline{C_p}(i, 0^\circ)$ and $\overline{C_p}(i, 0^\circ)$ profiles on the windward side of models with continuous appurtenances exhibit “zigzag variations” when $h_r > 4\%$. Fig. 7(b) and (d) show that the $\overline{C_p}(i, 0^\circ)$ of discontinuous appurtenance models are clearly larger than that of model A1 and the largest increment of 11.1% can be observed in model A4-III. While the $\overline{C_p}(i, 0^\circ)$ profiles for all types of discontinuous appurtenance model have similar pattern and values with model Ref. Fig. 8 shows the average mean and fluctuating pressure coefficients on the side façade. One can see that the $\overline{C_p}(i, 0^\circ)$ profiles on the upper half of side façade are close to the profile of model Ref when the appurtenances are continuous. But the absolute values of $\overline{C_p}(i, 0^\circ)$ on the lower half with no appurtenance increase obviously from model Ref. for the cases with discontinuous plates, the values of $\overline{C_p}(i, 0^\circ)$ in upper level can be clearly reduced around 12.4%, suggesting larger effects on wind load reduction on the side façade. The profiles of $\overline{C_p}(i, 0^\circ)$, however, seriously decrease for all appurtenance models with the maximum decrement of 17.5% found in the $\overline{C_p}(i, 0^\circ)$ profile of model A4-III. It can be concluded that the appurtenances arrangements adopted in this study are able to reduce the fluctuating wind pressures at each level, but with enlarged mean wind pressures on the lower part of the side façade. More details about the effect of façade appurtenances on the local peak pressure is referred (Yuan et al., 2018).

3.2. Results of along-wind layer force

In order to investigate the effect of appurtenances on the along-wind local force coefficients, the comparisons of along-wind layer force coefficients profile at direction of 0° for models with different h_r , b_r and d_r

are plotted in Figs. 9–11, respectively. In the model Ref, it can be found that the mean along-wind layer force coefficients increases with the model height with a maximum value of 0.98 at about $0.95H$ height, while the fluctuating along-wind layer force show the maximum value of 0.35 at that height.

Fig. 9 compares the $\overline{C_{pd}}(i, 0^\circ)$ and $\overline{C'_{pd}}(i, 0^\circ)$ of four models equipped with continuous appurtenances with those of model Ref. The dash dotted lines in the figures indicate the location of appurtenances. Fig. 9(a) and (b) show that the pattern of $\overline{C_{pd}}(i, 0^\circ)$ and $\overline{C'_{pd}}(i, 0^\circ)$ for model A1-III is similar to that for the smooth façade model, but the values of maximum $\overline{C_{pd}}(i, 0^\circ)$ and maximum $\overline{C'_{pd}}(i, 0^\circ)$ are larger by 3.9% and 5.1% respectively in model A1-III. It is also noted that the variation of $\overline{C_{pd}}(i, 0^\circ)$ with height becomes irregular with larger h_r with a clear “zigzag” shape. Further check on the variations shows that the value of $\overline{C_{pd}}(i, 0^\circ)$ sharply decreases when the test layer is located just under the appurtenance with small distances. The magnitudes of reduction of these layers are small in higher positions and become large when they are in lower storeys. Such results may indicate that the vertical flow along the windward façade induced by the downwash effect is suppressed by the splitter plates, and such effect of suppression becomes clearer at relatively lower position. Similar “zigzag variations” can also be observed from the fluctuating local force coefficients $\overline{C'_{pd}}(i, 0^\circ)$ of these continuous appurtenances models in Fig. 9(d)–(h). All these results are coincident with the multi-stagnation results observed on the windward side as shown in Fig. 5.

Figs. 10 and 11 compare the $\overline{C_{pd}}(i, 0^\circ)$ and $\overline{C'_{pd}}(i, 0^\circ)$ of the models equipped with discontinuous appurtenances with those of model Ref at wind direction of 0° . It can be observed that the values of $\overline{C_{pd}}(i, 0^\circ)$ of the models studied are all larger than those of model Ref. It can also be found that the discontinuous appurtenances models have the maximum values of $\overline{C_{pd}}(i, 0^\circ)$ at the height of $0.86H$ – $0.94H$ which is slightly lower than that of model Ref. For models with different b_r (Fig. 10(a)), the profiles of $\overline{C_{pd}}(i, 0^\circ)$ are noted almost identical when b_r varies from 20% to 60%, and

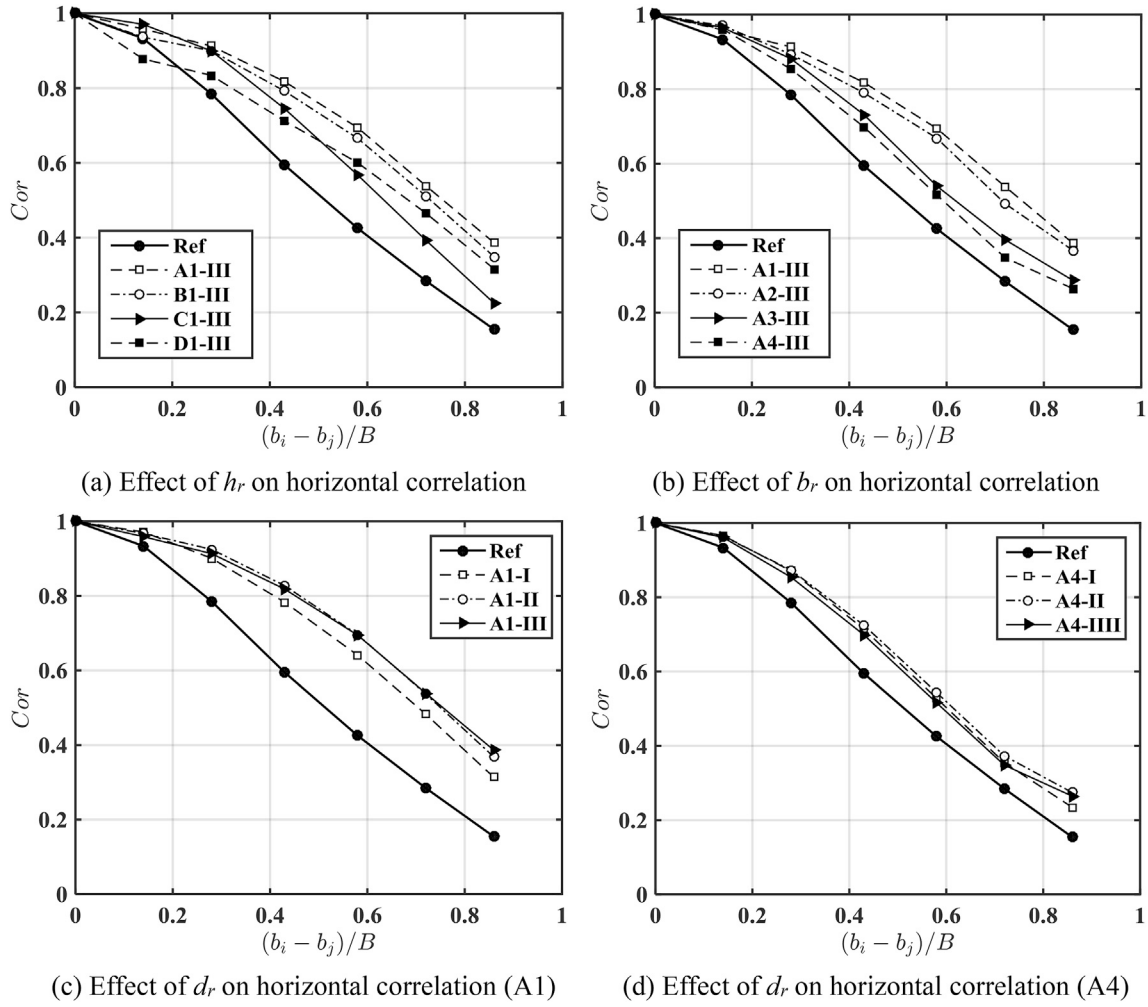


Fig. 16. Horizontal correlation of points on the side façade.

they are all slightly greater than the results of model Ref. For models with different d_r (Fig. 11(a)), the results of $\bar{C}_{pD}(i, 0^\circ)$ are quite similar, indicating that $\bar{C}_{pD}(i, 0^\circ)$ is not sensitive to the depth of plates, and a maximum increment of 5.8% in $\bar{C}_{pD}(i, 0^\circ)$ can be found in model A4-I. Figs. 10(b) and 11(b) show that the $\bar{C}_{pD}(i, 0^\circ)$ profiles of discontinuous appurtenance models have little difference with those of model Ref. These results agree with that shown Fig. 7, indicating that the discontinuous appurtenances lead to slightly larger along-wind mean layer forces than that from smooth façade, but have negligible effect on the fluctuating layer forces.

3.3. Results of cross-wind layer force

3.3.1. Distribution of cross-wind force coefficients

Since the mean cross-wind layer force coefficients $C'_{fL}(i, 0^\circ)$ at direction of 0° are very small and close to null, only the fluctuating values $C'_{fL}(i, 0^\circ)$ are plotted and discussed, as shown in Fig. 12. It can be observed that the $C'_{fL}(i, 0^\circ)$ of model Ref increases with model height reaching the maximum value at $0.7H$, and then decreases with height. The largest $C'_{fL}(i, 0^\circ)$ of model Ref is about 0.38. The profiles of fluctuating cross-wind layer force for continuous appurtenance models are similar to that of model Ref. The continuous appurtenance models show larger values at the layers higher than $0.7H$ and smaller values at lower height compared to the model Ref. These results are quite different with those shown in Fig. 8(c), which suggests that continuous appurtenances can reduce the average fluctuating local pressure, but increase the

fluctuating cross-wind layer forces at upper levels at the same time. Moreover, the cross-wind layer forces also show “zigzag variations” when h_r is larger than 4% similar to the profiles of along-wind layer force. The comparison of $C'_{fL}(i, 0^\circ)$ of the section within two vertically adjacent plates shown in Fig. 12(b)–(d) shows that $C'_{fL}(i, 0^\circ)$ at higher level of the section is smaller than that at lower level when the height of target section is lower than $0.7H$. However, such pattern becomes opposite when the height of target section is higher than $0.7H$.

Fig. 13 shows the fluctuating cross-wind layer force profile of models with different b_r . It is noted that the discontinuous appurtenances models exhibit smaller values of $C'_{fL}(i, 0^\circ)$ compared to model Ref in most layers especially at the height of $0.7H$. Moreover, the profiles of layer forces tend to decrease as b_r becomes larger. The peak value of $C'_{fL}(i, 0^\circ)$ 0.35 in model A4-III can be found at $0.7H$, indicating a decrement of 8.6% in the maximum fluctuating cross-wind layer force from that for model Ref. The layer forces at the height range of $[0.5H, 0.7H]$ almost keep constant when $b_r = 60\%$ (model A4-III), suggesting a better aerodynamic performance.

The effects of d_r on cross-wind layer force are also checked with model A4. The layer force profiles are shown in Fig. 14 with comparison to model Ref. It is noted that all three models have similar pattern of $C'_{fL}(i, 0^\circ)$ profiles and have smaller layer forces than that in model Ref, especially at the middle and upper part of the model height. Moreover, the $C'_{fL}(i, 0^\circ)$ can be reduced by about 3.9%, 7.0% and 8.6% in the A-I, A-II and A-III models respectively with the increase of d_r .

It is interesting to note that the appurtenances can significantly

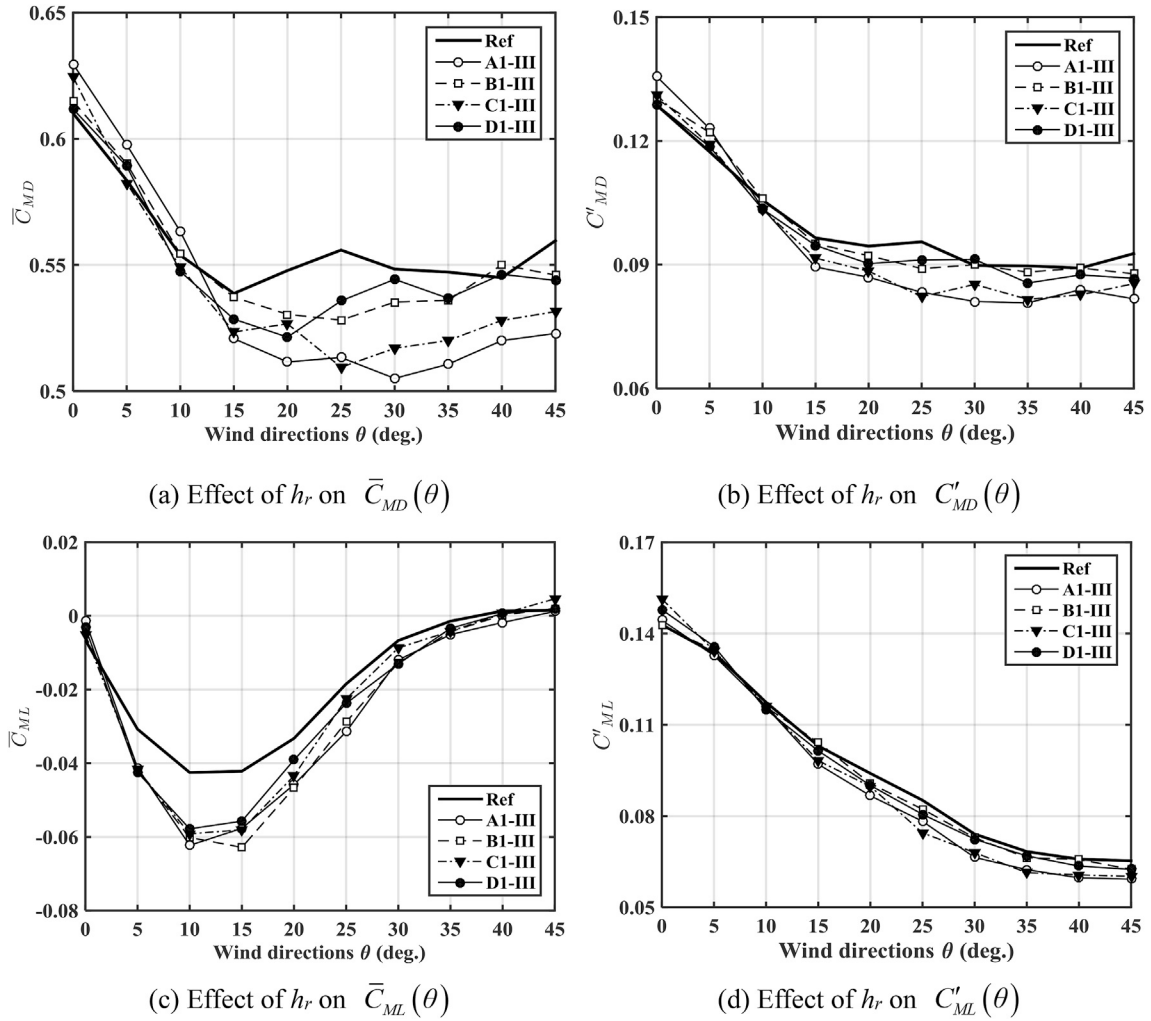


Fig. 17. Effect of vertical separation distance on base moment coefficients.

reduce the fluctuating pressures on the side facades of building for both the continuous and discontinuous arrangements as shown in Fig. 8(c) and (d). Such results, however, cannot be observed from the cross-wind fluctuating layer force. Although the discontinuous arrangements can result in a reduction of $C'_{fl}(i, 0^\circ)$ compared to the smooth model, the magnitude of reduction is much smaller than that in fluctuating pressure, and the continuous arrangement can even increase the $C'_{fl}(i, 0^\circ)$. The reason of this behavior will be further discussed below through analyzing the power spectrum and correlations of response.

3.3.2. Results of power spectra of cross-wind layer force

Fig. 15 presents the power spectra of cross-wind layer force at $0.7H$ when θ is 0° with largest cross-wind layer force on the model. All the power spectra are noted showing a dominant peak near the Strouhal number 0.1. This can be attributed to the fact that the basic cross-section of these test models is square. The existence of appurtenances does not change the shape of cross-section, so that the periods of vortex shedding of all models are not affected by the thin plates. The peaks of the power spectra of models equipped with appurtenances become slightly higher than that in model Ref, accompanying with narrower bandwidths. Such results indicate that the shed vortex can be better organized due to the appearance of appurtenances. Therefore, the correlations between different points are also checked below.

3.3.3. Result of horizontal correlation

It is known that if the correlation coefficients at the same level are

high, it will result in higher amplitude of layer force. In other words, the flow at this level near the wall is better formed with lower turbulence. The pressure points on the side face at the $0.7H$ height are selected to calculate the horizontal correlation coefficients of wind pressure as shown in Fig. 16. The point close to the leading edge is used as the reference. It is noted that the correlation coefficients of model Ref decrease from the value of 1.0 to 0.15 with increase of the distance between the points studied and the reference point, while the decreasing rates are much lower for models with appurtenances. Fig. 16(a) and (b) show that the decreasing trend of the correlation becomes steeper with increase in both b_r and h_r . Fig. 16(c) and (d) show that the decreasing rate of correlation is not sensitive to d_r for both continuous and discontinuous arrangement of appurtenances. These results show that the horizontal plates may result in a better organization of vortex shedding by restricting the flow in vertical direction. The results shown in Fig. 8 also show that the appurtenance can reduce the intensity of the vortex. The cross-wind layer forces might be either enlarged or reduced depending on the balance of these two effects. These results also explain the reason why model A4 has the best performance on the reduction of fluctuating cross-wind layer force in all models.

3.4. Results of base moment

3.4.1. Base moment coefficients and power spectrum

The variations of base moment coefficients with wind direction for models with different arrangements of appurtenance are shown in

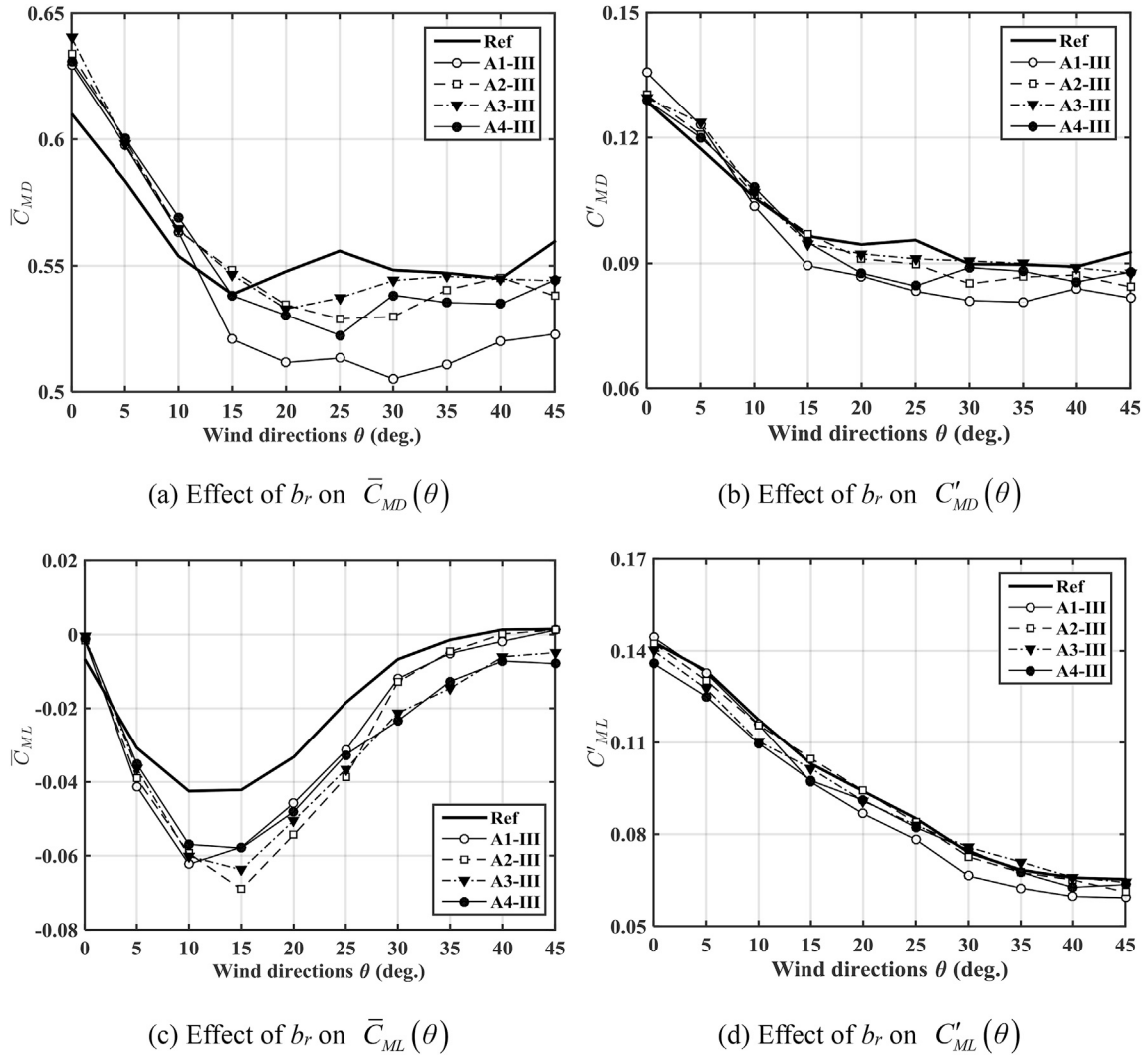


Fig. 18. Effect of horizontal gap on base moment coefficients.

Figs. 17 and 18. The maximum mean and fluctuating along-wind base moment coefficients for the smooth façade model are found to be 0.610 and 0.129 when the wind direction is 0° . The fluctuating cross-wind base moment coefficient also shows maximum value of 0.143 at wind direction of 0° . These observations are consistent with values and trends of coefficients variation for square-sectioned models in the literature (Lin et al., 2005; Tanaka et al., 2012).

To study the effect of vertical separation of appurtenances four models with $d_r = 12.5\%$ equipped with continuous splitter plates are compared with model Ref and the results of along-wind base moment coefficients are shown in Fig. 17(a) and (b). The $|\bar{C}_{MD}(\theta)|_{max}$ and $|C'_{MD}(\theta)|_{max}$ of these models are for 0° wind direction. It is noted that the along-wind base moments of test models are larger than that of model Ref when $\theta \in [0^\circ, 15^\circ]$, and they become much smaller with an increase of θ . The maximum decrement of 7.9% from model Ref can be found in $\bar{C}_{MD}(\theta)$ of model A1-III when $\theta = 30^\circ$. Fig. 17(c) shows that the models give values of $\bar{C}_{ML}(\theta)$ closed to null at 0° and 45° while the maximum $\bar{C}_{ML}(\theta)$ can be found for $\theta = 15^\circ$. The models with appurtenance have larger values than the smooth façade model. Fig. 17(d) shows that the $C'_{ML}(\theta)$ of model type of III with appurtenance have same variation with model Ref, but the values are slightly smaller when $\theta \geq 10^\circ$.

The results of four appurtenances models, with same h_r and d_r but different b_r , are shown in Fig. 18. Model A1-III is equipped with continuous appurtenances while the other three models are equipped

with discontinuous appurtenances. The reduction effect on the along-wind base moment at larger θ can also be found in discontinuous appurtenance models, but it is much weaker compared to that in model A1-III with continuous appurtenances. The discontinuous appurtenance models show similar pattern of $\bar{C}_{MD}(\theta)$ and $C'_{ML}(\theta)$ with model Ref and A1 as shown in Fig. 18(c) and (d).

The maximum base moment coefficients of each model from all wind directions are shown in Fig. 19, where the Y-axis denotes the ratio of maximum base moment coefficient of a model to the value of the model Ref. Both Fig. 19(a) and (b) show that the variation of along-wind base moment have the same pattern when $d_r > 7.5\%$. One can see that the ratios for model B1 and D1 are close to 1.0, suggesting much lower values compared to models A1 and C1. Such results indicate that the along-wind base moment of appurtenances model is almost equal to the moment of the smooth façade model when vertical separation ratio is at 8% or 16%. When $d_r = 7.5\%$, however, the maximum $|\bar{C}_{MD}(\theta)|_{max}$, $|C'_{MD}(\theta)|_{max}$ and $|C'_{ML}(\theta)|_{max}$ can increase by 5%, 7% and 7.4% compared with model Ref. It can also be found in Fig. 19(c) that $|C'_{ML}(\theta)|_{max}$ becomes smaller when $d_r > 7.5\%$, especially for models A1 and B1. In addition, the maximum $|C'_{ML}(\theta)|_{max}$ can be increased by 6% when $h_r \in [12\%, 16\%]$, but close to that of the smooth façade model when h_r is smaller. Such results show that the continuous appurtenance with relative large h_r and small d_r may increase the cross-wind base moment. Further check on the results of discontinuous appurtenance models in Fig. 19(c) shows that the largest

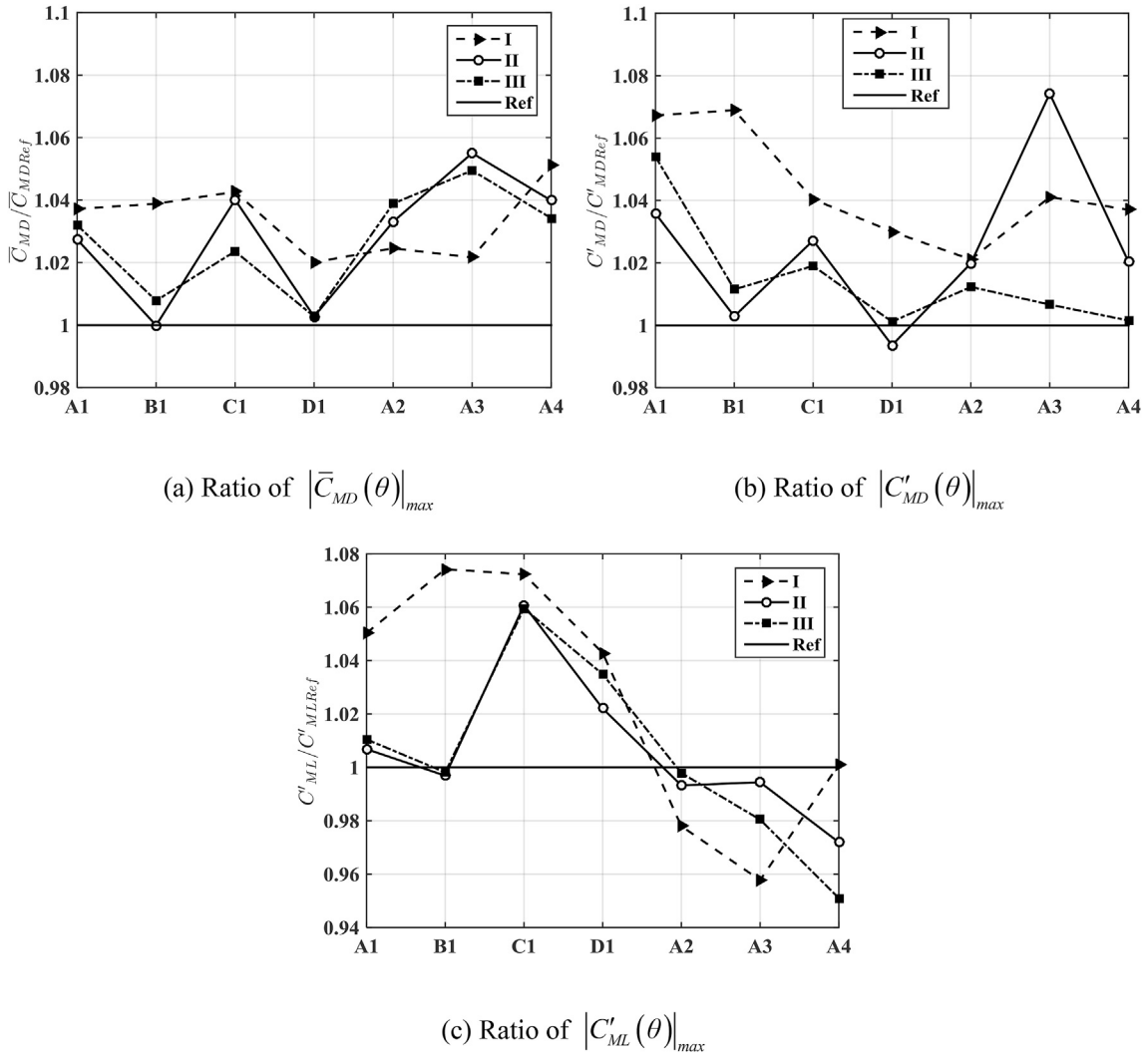


Fig. 19. The ratio of maximum base moment coefficient.

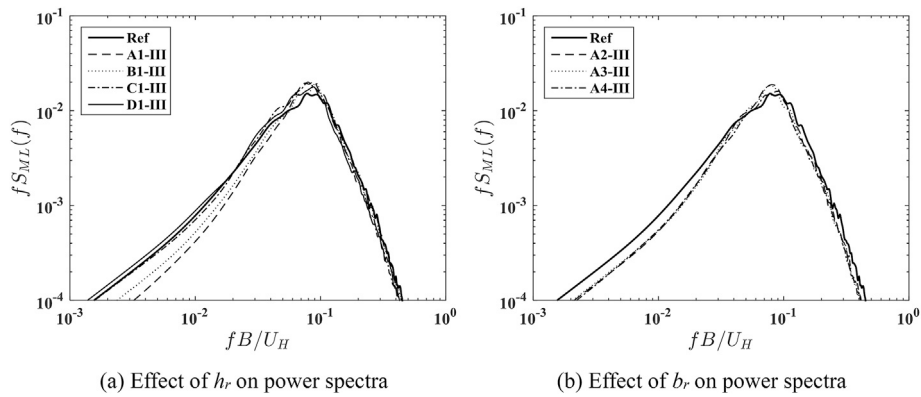


Fig. 20. Power spectra of cross-wind base moment.

decrement of $|C'_{ML}(\theta)|_{max}$ can reach 5% for model A4-III. When $d_r > 7.5\%$, it can be found that the cross-wind base moment decreases with an increase of b_r .

Fig. 20 shows the power spectra of cross-wind base moment for model Ref and the test models when $\theta = 0^\circ$. The dominant peaks of power spectra for all models are found close to 0.1, while the power spectra of models equipped with appurtenances show larger peak values and

narrower bandwidths. Such results are quite similar to those observed in Fig. 15.

3.4.2. Results of vertical correlation of layer forces

The correlation of layer forces is studied with an attempt to find answers to the difference of base moments from that of the Model Ref. The top layer is selected as the reference layer. The correlations of the along-

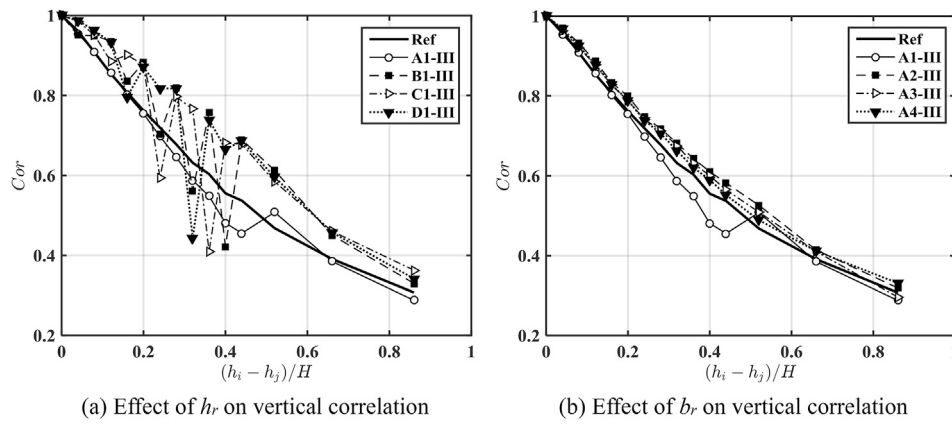


Fig. 21. Vertical correlation of along-wind layer forces.

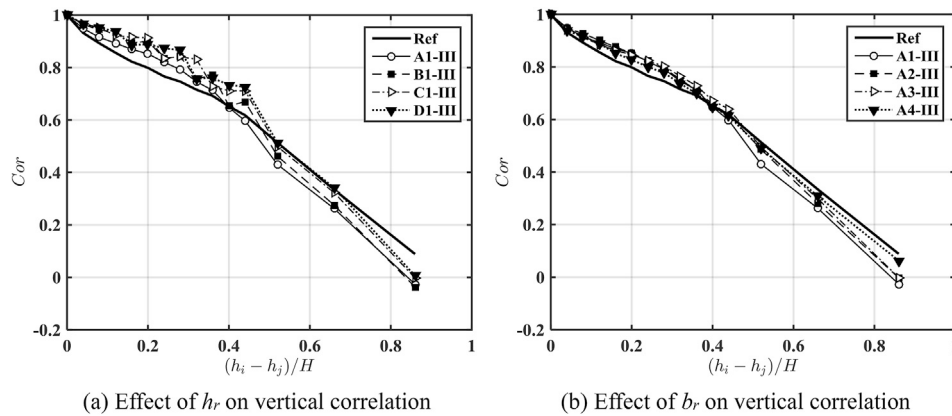


Fig. 22. Vertical correlation of cross-wind layer forces.

wind and cross-wind layer forces at wind direction of 0° are respectively shown in Figs. 21 and 22.

Fig. 21 shows the “zigzag variations” in the along-wind vertical correlation profiles of models with continuous appurtenance. The correlation of discontinuous appurtenance model becomes smaller with an increase of b_r , indicating that the larger horizontal gaps between plates are able to decrease the correlation along the height on the windward side. The appurtenance models in Fig. 22 show larger cross-wind vertical correlation at the higher part of the façade. Smaller values are found at the lowest three layers where there is no appurtenances. This indicates that the appurtenances on the façade can increase the correlation between layer forces along the height. Comparing Fig. 22(a) and (b) shows that when the appurtenances are continuous, the correlations between different layers tend to be larger than that with discontinuous appurtenances. This also explains the reason why the discontinuous type appurtenances can slightly decrease the overall cross-wind fluctuating load.

4. Conclusions

A series of wind tunnel tests have been conducted to investigate the effect of different appurtenances configurations on the aerodynamic forces of high-rise buildings, including the mean and fluctuating base moment, layer force, power spectral densities, correlation and coherence etc. The following conclusions can be obtained from this study.

- (1) A “zigzag variation”, which is closely related to the location of appurtenances, is found in both along-wind and cross-wind layer forces when the vertical separation distance ratio h_r of continuous appurtenances is more than 4%. Besides, the discontinuous appurtenances could slightly increase the along-wind layer force, but

severely reduce the cross-wind layer force. The reduction effect on the cross-wind layer force is stronger with an increase of b_r and d_r . The decrement of $C'_{fl}(i, 0^\circ)$ even up to 8.6% at the height of $0.7H$ for model A4-III.

- (2) The results of power spectra indicate that the shed vortex can be better organized due to the presence of appurtenances, and this is quite different from the observed results of fluctuating pressures.
- (3) The results of correlation between the pressure taps at the same level indicate that the horizontal plates may result in a better organization of vortex shedding by restricting the flow in vertical direction and reduce the intensity of the vortex. The cross-wind layer forces might therefore be either enlarged or reduced depending on the balance of the two effects.
- (4) The façade appurtenances, regardless of type, could slightly increase the base moment in the along-wind direction. But for the cross-wind direction, the appurtenance type results in very different effect on the base moment. The continuous appurtenances could increase the cross-wind base moment and this effect becomes stronger for models with $h_r = 12\%$ than other models.
- (5) Because of the trade-off between the reduction of local wind load and the enhancement of wind load correlation between different positions (in both horizontal and vertical directions), the overall wind loads of building with façade appurtenances show only slight change from the smooth surface building. The discontinuous appurtenances reduces the fluctuating cross-wind base moment with the largest decrement up to 5% for model A4-III ($h_r = 4\%$, $d_r = 12.5\%$), while the continuous plates can increase this value up to 8%. Such a variation might be negligible from the engineering point of view considering the uncertainty of

experiment, the reason of this difference is important for understanding of the mechanism of wind load.

Acknowledgement

This study was supported in part by Chongqing Science and Technology Bureau (cstc2018jcyj-yszxX0010), the Fundamental Research Funds for the Central Universities (Grant No. 2019CDXYTM0032) and the 111 project of the Ministry of Education and the Bureau of Foreign Experts of China (No. B18062). The efforts and comments of Prof. S.S. Law in improving the English of this paper are gratefully acknowledged.

References

- Carassale, L., Freda, A., Marrè-Brunenghi, M., 2014. Experimental investigation on the aerodynamic behavior of square cylinders with rounded corners. *J. Fluids Struct.* 44 (7), 195–204.
- Chand, I., Bhargava, P.K., Krishak, N.L.V., 1998. Effect of balconies on ventilation inducing aero motive force on low-rise buildings. *Build. Environ.* 33 (6), 385–396.
- GB50009-2012, 2012. Load Code for the Design of Building Structures. China Architecture & Building Press, Beijing.
- Hui, Y., Tamura, Y., Yang, Q.S., 2017. Analysis of interference effects on torsional moment between two high-rise buildings based on pressure and flow field measurement. *J. Wind Eng. Ind. Aerodyn.* 164, 54–68.
- Kumar, A., Rahul, P.S., Kumar, S., 2013. Performance optimization of tall buildings subjected to wind-an Indian scenario. In: Eighth Asia-Pacific Conference on Wind Engineering. Chennai, India, pp. 817–826.
- Kwok, K.C.S., Wilhelm, P.A., Wilkie, B.G., 1988. Effect of edge configuration on wind-induced response of tall buildings. *Eng. Struct.* 10 (2), 135–140.
- Letchford, C.W., Lander, D.C., Case, P., Dyson, A., Amitay, M., 2016. Bio-mimicry inspired tall buildings: the response of cactus-like buildings to wind action at Reynolds Number of 10^4 . *J. Wind Eng. Ind. Aerodyn.* 150, 22–30.
- Li, Y., Tian, X., Tee, K.F., Li, Q.S., Li, Y.G., 2018. Aerodynamic treatments for reduction of wind loads on high-rise buildings. *J. Wind Eng. Ind. Aerodyn.* 172, 107–115.
- Lin, N., Letchford, C., Tamura, Y., Liang, B., Nakamura, O., 2005. Characteristics of wind forces acting on tall buildings. *J. Wind Eng. Ind. Aerodyn.* 93 (3), 217–242.
- Maruta, E., Kanda, M., Sato, J., 1998. Effects on surface roughness for wind pressure on glass and cladding of buildings. *J. Wind Eng. Ind. Aerodyn.* 74–76, 651–663.
- Quan, Y., Kuang, J., Gu, M., Wang, S., 2016. Effects of grid curtains on the wind loads of a high-rise building. *Struct. Des. Tall Special Build.* 25 (5), 245–262.
- Quan, Y., Hou, F., Gu, M., 2017. Effects of vertical ribs protruding from facades on the wind loads of super high-rise buildings. *Wind Struct.* 24 (2), 145–169.
- Stathopoulos, T., Zhu, X., 1988. Wind pressure on building with appurtenances. *J. Wind Eng. Ind. Aerodyn.* 31, 265–281.
- Stathopoulos, T., Zhu, X., 1990. Wind pressure on building with mullions. *J. Struct. Eng.* 116 (8), 2272–2291.
- Tamura, Y., Xu, X., Tanaka, H., Kim, Y.C., Yoshida, A., Yang, Q., 2017. Aerodynamic and pedestrian-level wind characteristics of super-tall buildings with various configurations. *Procedia Eng.* 199, 28–37.
- Tanaka, H., Tamura, Y., Ohtaka, K., Nakai, M., Kim, Y.C., 2012. Experimental investigation of aerodynamic forces and wind pressures acting on tall buildings with various unconventional configurations. *J. Wind Eng. Ind. Aerodyn.* 107–108, 179–191.
- Yu, X., Xie, Z., Gu, M., 2018. Interference effects between two tall buildings with different section sizes on wind-induced acceleration. *J. Wind Eng. Ind. Aerodyn.* 182, 16–26.
- Yuan, K., Hui, Y., Chen, Z., 2018. Effects of facade appurtenances on the local pressure of high-rise building. *J. Wind Eng. Ind. Aerodyn.* 178, 26–37.